Discrete Distributions

Binom(n, p)

Parameter:	$n \in \mathbb{N}, p \in [0, 1]$.
PMF:	$\mathrm{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0, \ldots, n.$
Moments:	$\mathrm{E}(X) = np$, $\mathrm{Var}(X) = np(1-p)$.
Motivation:	Describes the number of successes in n independent trials with two possible outcomes each.

Unif(a, b)

Parameter:	$a, b \in \mathbb{Z}, a \leq b$, $n = b - a + 1$.
PMF:	$\mathrm{P}(X = k) = 1/n, \qquad k = a, a+1, \ldots, b-1, b.$
Moments:	$\mathrm{E}(X) = (a+b)/2$, $\mathrm{Var}(X) = ((b-a+1)^2 - 1)/2$.
Motivation:	Describes a random variable whose outcomes are equally likely.

Pois(λ)

Parameter:	$\lambda \in \mathbb{R}, \lambda > 0$.
PMF:	$\mathrm{P}(X = k) = \lambda^k e^{-\lambda}/k!, \qquad k = 0, 1, 2, \ldots.$
Moments:	$\mathrm{E}(X) = \lambda$, $\mathrm{Var}(X) = \lambda$.
Motivation:	Describes the number of successes which occur when the probability of each success p is small, but the number of trials is large, i.e. $p \approx 0$ but $np = \lambda$.

Geo(p)

Parameter:	$p \in (0, 1)$.
PMF:	$\mathrm{P}(X = k) = (1-p)^{k-1} p, \qquad k = 1, 2, \ldots.$
Moments:	$\mathrm{E}(X) = 1/p$, $\mathrm{Var}(X) = (1-p)/p^2$.
Motivation:	Describes the number of failed independent trials with success probability p until the first success.

Continuous Distributions

Beta(a, b)

Parameter:	$a, b > 0$.
Density:	$f(x) = x^{a-1}(1-x)^{b-1}/B(a, b), \qquad 0 \leq x \leq 1.$
Moments:	$\mathrm{E}(X) = a/(a+b)$, $\mathrm{Var}(X) = \frac{ab}{(a+b)^2(a+b+1)}$.
Motivation:	Canonical density on $[0, 1]$, reduces to the uniform density when $a = b = 1$.

$\mathbf{N}(\mu, \sigma^2)$

Parameter:	$\mu, \sigma^2 \in \mathbb{R}, \sigma^2 > 0.$
Density:	$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \qquad x \in \mathbb{R}.$
Moments:	$\mathrm{E}(X) = \mu$, $\mathrm{Var}(X) = \sigma^2.$
Motivation:	Appears as the limiting distribution in the central limit theorem.

$\mathbf{Exp}(\lambda)$

Parameter:	$\lambda \in \mathbb{R}, \lambda > 0.$
Density:	$f(x) = \lambda e^{-\lambda x}, \qquad x > 0.$
Moments:	$\mathrm{E}(X) = 1/\lambda$, $\mathrm{Var}(X) = 1/\lambda^2.$
Motivation:	Often used to model the length of random time intervals and waiting times until an event.

Complicated Distributions

You should be aware of these, but need not remember pdfs or moments in detail.

$\mathbf{Hypergeo}(n, K, m)$

Parameter:	$n \in \mathbb{N}, K \;\&\; m \in \{0, \ldots, n\}.$
PMF:	$\mathrm{P}(X = k) = \binom{K}{k}\binom{n-K}{m-k}/\binom{n}{K}, \qquad k = 0, \ldots, \min\{n, K\}.$
Moments:	$\mathrm{E}(X) = np$, $\mathrm{Var}(X) = np(1-p).$
Motivation:	Describes the number of successes in m independent trials sampled without replacement from a population of size n containing K successes.

$\mathbf{t}(n)$

Parameter:	$n \in \mathbb{N}.$
Density:	$f(x) = \frac{\Gamma(\frac{n+1}{2})}{\sqrt{n\pi}\Gamma(\frac{n}{2})} \left(1 + \frac{x^2}{n}\right)^{-(n+1)/2}, \qquad x \in \mathbb{R}.$
Moments:	$\mathrm{E}(X) = 0$ (for $n > 1$), $\mathrm{Var}(X) = \frac{n}{n-2}$ (for $n > 2$).
Motivation:	Commonly used for hypothesis tests and the calculation of confidence intervals.

$\boldsymbol{\chi}^{\mathbf{2}}(n)$

Parameter:	$n \in \mathbb{N}.$
Density:	$f(x) = \frac{1}{2^{n/2}\Gamma(\frac{n}{2})} x^{n/2-1} e^{-x/2}, \qquad x > 0.$
Moments:	$\mathrm{E}(X) = n$, $\mathrm{Var}(X) = 2n.$
Motivation:	Describes the sum of n independent squared standard normal random variables. Often arises as a test statistic.

Gamma(α, β)	
Parameter:	$\alpha, \beta > 0$.
Density:	$f(x) = \beta^\alpha x^{\alpha-1} e^{-\beta x}/\Gamma(\alpha), \qquad x > 0.$
Moments:	$\mathrm{E}(X) = \alpha/\beta$, $\mathrm{Var}(X) = \alpha/\beta^2$.
Motivation:	Sum of α independent $\mathrm{Exp}(\beta)$-random variables when α is an integer.